

Muhammad Daffa' Fisabilillah

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Summary

Software Engineer with strong foundations in computer engineering, mobile development, and applied machine learning. Experienced in building end-to-end applications, from designing modular architectures to integrating APIs and deploying production-ready solutions. Skilled in Android (Kotlin, Jetpack, Compose) and familiar with backend and cloud services, with additional expertise in machine learning (TensorFlow Lite, PyTorch) for real-time processing. Adept at working in cross-functional teams, leading projects, and delivering impactful solutions across mobile, web, and AI domains.

Education

Institut Teknologi Sepuluh Nopember , Computer Engineering	Sept 2021 – Sept 2025
• GPA: 3.53/4.0	
MAN Insan Cendekia OKI , Science	Sept 2018 – Sept 2021
• GPA: 92/100	

Experience

Mobile Android Developer , PT. Adinata Charming Emmanuel – Surabaya, Indonesia (Freelance)	Jul 2024 – Feb 2025
- Developed Ace Member, a membership & promo management app for 4 user roles (Admin, Mitra, Resepsionis, Member). - Implemented 20+ REST API integrations (Auth, Promo, Merchant, File Upload, Token) using Kotlin and Clean Architecture (MVVM). - Built modular XML-based UI from Figma Design System covering authentication, dashboard, coupon, and membership flows. - Conducted 40+ end-to-end test scenarios across multiple roles, ensuring feature compliance and reducing bugs before release. - Delivered MVP within project timeline using Scrum (~90% sprint completion across 10+ sprints).	
Mobile Application Developer (Study Independet) , Bangkit Academy – Remote (Indonesia)	Feb 2024 – Aug 2024
- Graduated from Bangkit Academy 2024 (Google, GoTo, Traveloka) with final score 89.08/100 (Grade A) and 100% attendance. - Achieved A grade in Capstone Project (91.6/100), contributing as Mobile Developer in a 6-person cross-functional team (2 Cloud, 2 ML, 2 Mobile) building <i>TerraVision</i>, a real estate investment platform. - Built Android apps using Kotlin, Jetpack Compose, Room, and MVVM, integrating REST APIs, local databases, and on-device TensorFlow Lite models. - Completed elective certifications: <i>Belajar Dasar UX Design</i>, <i>Jetpack Compose</i>, and <i>Android Developer Expert</i>.	

Projects

BAKI: Sport Motion Detector Based on ML Kit in Mobile App

- Led a 3-member team (Mobile, ML, backend) in developing **BAKI**, an Android fitness app with AI-powered form correction, built under Scrum with 25 tasks and ~90% completion.
- Built core engine for **real-time workout classification** (push-up, sit-up, squat, lunges) using ML Kit pose landmarks (33 points), KNN classifier, and EMA smoothing; integrated with automatic repetition counter.
- Delivered supporting features: OTP-based authentication, Daily Plan scheduling, workout history, and auto rest timer.

- Developed performance modules: **Anticipation Test** (ball-drop timing) and **Reflex Test** (audio-based, 6-level benchmark).
- Built 40+ XML layouts with Material Design, CameraX live tracking, and Clean Architecture (MVVM + Dagger Hilt), ensuring modularity and maintainability.

DogVision: On-Device Dog Breed Classifier Using PyTorch and TFLite

- Built **custom CNN with residual blocks** from scratch (PyTorch) and applied Mixup augmentation on 10-breed Stanford Dog dataset (20K images).
- Achieved **70% validation accuracy (scratch)** and up to **85–86% with transfer learning** (MobileNetV2 on mobile, ResNetV2 on web).
- Deployed models cross-platform: PyTorch → ONNX → TensorFlow → TFLite for Android (offline inference) and Flask + TensorFlow for web.
- Delivered Android app with **real-time image classification** and local history (Room DB) for offline usage.

Digital Twin for Virtual Music Concert Using Metahuman UE5

- Developed a photorealistic avatar for a virtual musician using Unreal Engine 5 and Metahuman, enabling immersive concert performance in a 3D environment
- Integrated motion capture systems (Move AI, Rokoko, DeepMotion), facial tracking (Live Link Face, Face Landmarker Link 02), and clothing simulation (Marvelous Designer)
- Achieved natural motion and facial expression synchronization with real-time rendering performance at 1080p 30 FPS
- Evaluated avatar realism and blendshape precision using custom analysis pipeline with MediaPipe for eye and mouth landmarks

Organizations

Human Resources Development Staff, HIMATEKKOM – ITS

Apr 2023 - Mar 2024

- Contributed to HR development programs, focusing on member growth, engagement, and appreciation initiatives.
- Supported major events such as **OKKBK 2023** (student orientation, 100+ freshmen) and **LKMM-TD III** (leadership training, 36 participants, 90% attendance).
- Contributed to the **BUDAPEST Program**, collecting and recognizing student achievements across entrepreneurship, managerial, academic, and talent domains.

Head of Event Organizer Subdivision, MAGE 9 – ITS

Feb 2023 – Nov 2023

- Led the Event Organizer Subdivision to plan and execute national-scale events (Workshops, Opening Ceremony, Main Event, and Final Exhibition) with 190+ participants.
- Coordinated across 10+ cross-functional divisions (Logistics, Public Relations, Sponsorship, Multimedia, Competitions) to ensure integrated event management.

Staff of Computer Olympiad Division, MAGE 8 – ITS

Jun 2022 – Dec 2022

- Assisted in preparing **competition guidelines and 60+ problem sets** (Physics, Programming, Logic) for 40+ participants.
- Improved participant engagement by proposing better archiving, web tutorials, multi-channel communication and technical meetings (TM).

Technologies

Languages: Kotlin, Java, Python, XML, SQL

Frameworks & Libraries: Jetpack Compose, Android Jetpack (Navigation, Room, ViewModel, LiveData), Dagger Hilt, PyTorch, TensorFlow Lite, OpenCV, Keras

Tools & Platforms: Android Studio, Firebase, Git, CameraX, ML Kit, ONNX, CSV Processing, Gradle, GitHub, Jupyter Notebook

Concepts: Clean Architecture (MVVM), Agile Development (Scrum), On-Device Machine Learning, Real-Time Image Processing, Pose Estimation, Mobile Deployment Optimization, REST API Integration, Material Design